

Graphing Linear Functions – Intercepts and Slope

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A **linear function** is an equation of the form

$$y = mx + b,$$

where m is the **slope** and b is the **y-intercept**.

X-Intercept

The **x-intercept** is the point where the graph crosses the **x-axis**.

At the x-intercept,

$$y = 0.$$

Y-Intercept

The **y-intercept** is the point where the graph crosses the **y-axis**.

At the y-intercept,

$$x = 0.$$

Graphing a Linear Function

To graph a **linear function**:

Plot the **y-intercept**.

Use the **slope** to find another point.

Draw the line through the points.

Label both **intercepts** clearly.

Examples

Example 1

Graph

$$y = \frac{1}{4}x - 1$$

and determine its intercepts and slope.

Example 1

The slope is $m = \frac{1}{4}$ and the y-intercept is $(0, -1)$.

For the x-intercept, let $y = 0$:

$$0 = \frac{1}{4}x - 1 \implies x = 4.$$

Therefore, x-intercept $(4, 0)$,
y-intercept $(0, -1)$.

$$\boxed{m = \frac{1}{4}}$$

Example 2

Graph $y = \frac{2}{3}x - 2$ and determine its intercepts and slope.

Example 2

The slope is $m = \frac{2}{3}$ and the y-intercept is $(0, -2)$.

For the x-intercept:

$$0 = \frac{2}{3}x - 2 \implies x = 3.$$

Therefore, x-intercept $(3, 0)$,
y-intercept $(0, -2)$.

$$\boxed{m = \frac{2}{3}}$$

Example 3

Graph

$$y = x - 3$$

and determine its intercepts and slope.

Example 3

The slope is $m = 1$ and the y-intercept is $(0, -3)$.

For the x-intercept:

$$0 = x - 3 \implies x = 3.$$

Therefore, x-intercept $(3, 0)$,
y-intercept $(0, -3)$.

$$\boxed{m = 1}$$

Practice Exercises

Practice 1

Graph each linear function and determine its **intercepts** and **slope**.

$$y = \frac{1}{4}x - 1$$

Practice 2

Graph each linear function and determine its **intercepts** and **slope**.

$$y = \frac{2}{3}x - 2$$

Practice 3

Graph each linear function and determine its **intercepts** and **slope**.

$$y = x - 3$$

Practice 4

Graph each linear function and determine its **intercepts** and **slope**.

$$y = 4x - 4$$

Practice 5

Graph each linear function and determine its **intercepts** and **slope**.

$$y = \frac{1}{4}x + 1$$

Practice 6

Graph each linear function and determine its **intercepts** and **slope**.

$$y = \frac{2}{3}x + 2$$

Practice 7

Graph each linear function and determine its **intercepts** and **slope**.

$$y = -\frac{3}{2}x + 3$$

Practice 8

Graph each linear function and determine its **intercepts** and **slope**.

$$y = 4x + 4$$

Practice 9

Graph each linear function and determine its **intercepts** and **slope**.

$$y = -\frac{1}{4}x - 1$$

Practice 10

Graph each linear function and determine its **intercepts** and **slope**.

$$y = \frac{2}{3}x - 1$$

Activities

Activity 1

Graph each linear function and determine its **intercepts** and **slope**.

$$y = -4x + 4$$

Activity 2

Graph each linear function and determine its **intercepts** and **slope**.

$$y = -\frac{3}{2}x + 3$$

Activity 3

Graph each linear function and determine its **intercepts** and **slope**.

$$y = -x + 3$$

Activity 4

Graph each linear function and determine its **intercepts** and **slope**.

$$y = -\frac{1}{4}x + 1$$

Activity 5

Graph each linear function and determine its **intercepts** and **slope**.

$$y = \frac{4}{5}x - 4$$

Activity 6

Graph each linear function and determine its **intercepts** and **slope**.

$$y = -3x - 3$$

Activity 7

Graph each linear function and determine its **intercepts** and **slope**.

$$y = -x - 2$$

Activity 8

Graph each linear function and determine its **intercepts** and **slope**.

$$y = -\frac{1}{3}x - 1$$

Activity 9

Graph each linear function and determine its **intercepts** and **slope**.

$$y = x + 4$$

Activity 10

Graph each linear function and determine its **intercepts** and **slope**.

$$y = \frac{3}{5}x - 3$$

Thank you!